



National Accreditation Board for Testing and Calibration Laboratories (NABL)

Assessment Forms and Checklist for Recognition of Temporary Site Laboratories Testing Aggregates and Concrete in Building Projects (Based on NABL 224)

**ISSUE NO.: 01
ISSUE DATE: 01.04.2024**

**AMENDMENT NO.:
AMENDMENT DATE:**

AMENDMENT SHEET

Sl.	Amendment No.	Page No.	Clause No.	Date of Amendment	Amendment	Reasons	Signature QA Team	Signature Competent authority
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National Accreditation Board for Testing and Calibration Laboratories

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ASSESSMENT SCHEDULE
(Temporary Site Laboratory)

Name of the Laboratory:		
Authorized representative of Laboratory:		Date of Visit:
Type of Assessment: Initial assessment / Surveillance / Supplementary Visit		
Field: Testing	Discipline (s): Mechanical	
Assessment Criteria: NABL 224		
Assessment Timings	Opening/Closing Meeting Date/Time	
<i>Morning: AM to PM</i> <i>Afternoon: PM to PM</i>	<i>Opening Meeting:</i> <i>Closing Meeting:</i>	
Name and Expertise of the Assessor	Schedule of Activity to be Assessed	
	Morning	Afternoon
Lead Assessor		
Assessor 1		
Assessor 2		
Observer (only for observation)		
I declare that the assessment schedule has been prepared after discussing with the assessment team members.		
Name & Signature of Lead Assessor		

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DETAILS OF TESTING / RE-TESTING / REPLICATE WITNESSED DURING ASSESSMENT
(Temporary site laboratory)

Name of the Laboratory:	
Discipline: Mechanical	Date of Assessment:
Materials or Products tested	
Specific Test Performed/ Tests or type of tests performed	
Test Method	
Range of Testing	
Measurement Uncertainty ($\pm$) at Value	

1. Fresh Sample

Sample ID	
Tested by	
Results of Test Witnessed	

2. Re-testing/replicate testing of Retained Sample

Sample ID	
Date of Earlier Testing	
Earlier Tested by	
Earlier Reported Results	
Results of Test Witnessed	
Test conducted by (same person)	
Test conducted by (different person)	

Remarks:

Deviations observed, if any	
Conclusion on the technical competence of the lab for the test witnessed	

(Enclose/upload all supporting data sheets for tests witnessed)

(Signature & Name of Assessor)

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**Summary on Non-Conformities
(Temporary site laboratory)**

(Please use separate sheet for raising each Non-Conformity)

Name of the Laboratory			
Name of the Assessor			
Date of Assessment		Discipline: Mechanical	
NON-CONFORMITY STATEMENT:			
Cl. No. of NABL 224:			
Acceptance (Signature of laboratory's representative and propose the required corrective actions)		Signature & Name of Assessor	
CORRECTION PROPOSED BY THE LABORATORY			
Signature & Name Authorized Representative of Laboratory			
ASSESSOR'S COMMENTS ON CORRECTIVE ACTION PROPOSED BY THE LABORATORY: (Root cause analysis to be submitted along with the evidences of corrective actions)			
Acceptance/Signature of Assessor			
REMARKS BY LEAD ASSESSOR, IF ANY:			
Acceptance/Signature & Name of Lead Assessor			

SUMMARY OF ASSESSMENT (Temporary Site Laboratory)

Sl.	Item	Details												
1.	Name of the Laboratory													
2.	Project Name													
3.	Project Size													
4.	Project duration (dd/mm/yyyy)	From: To:												
5.	Date of Assessment													
6.	Field													
7.	Discipline	Mechanical												
8.	Type of Assessment	Initial Assessment / Surveillance / Supplementary Visit												
9.	Assessment Team / Capacity	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr><td style="width: 50%; height: 20px;"></td><td style="width: 50%; height: 20px;"></td></tr> <tr><td style="height: 20px;"></td><td style="height: 20px;"></td></tr> <tr><td style="height: 20px;"></td><td style="height: 20px;"></td></tr> <tr><td style="height: 20px;"></td><td style="height: 20px;"></td></tr> <tr><td style="height: 20px;"></td><td style="height: 20px;"></td></tr> <tr><td style="height: 20px;"></td><td style="height: 20px;"></td></tr> </table>												
10.	Date of earlier visit:	<i>Non-Conformities during earlier visit have/ have not been discharged. (Yes or No)</i>												
11.	Total no. of Non-conformities													
12.	Assessment Summary													
13.	Recommendation of Assessment team as per NABL 224													
14.	Only if recognition is recommended, date by which the Corrective Action to be submitted by the lab for the above Non-conformities (Max. 30 days):													
Enclosures	NRF 6	NRF 4												
	NRF 3	NRF 1												
	NRF 1A	Checklist												
	NRF 2	NRF 72 (Scope)												
	NRF 74	TA/DA Forms												
	PT/ILC records	Any other docs.												
No. of Pages														
Acknowledgement by Authorised Representative of lab & Date		Signature of Lead Assessor& Date												

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ASSESSOR CHECK LIST (Temporary Site Laboratory)

Date of Assessment			
Type of Assessment	Initial assessment / Surveillance / Supplementary assessment		
Project Name and Address			
Name of the Laboratory			
Head of the Laboratory Name			
Name of the Assessor			
Clause No.	Requirements and Comments <i>Compliance = C, Non-compliance = NC</i>	C/NC	Remarks/Comments
5.0	Manpower Requirements		
5.1	All personnel of the laboratory shall be competent and work impartially.		
5.2	The laboratory shall have records for the competency of personnel (education, qualification, training, technical knowledge, skills and experience).		
5.3	The laboratory shall authorize personnel to perform testing activities in the laboratory.		
5.4	The laboratory shall ensure the training for newly recruited personnel and records to be maintained.		
5.5	All Site testing parameters shall be performed by the laboratory staff only. However, assistance/support staff for the operation/functioning of testing activities can be contractual/outsourced.		
5.6	Adequate testing personnel shall be available (depending upon the project size) to ensure quality control and assurance.		
5.7	The temporary site laboratory shall have personnel with the qualifications and experience as given in the table under Cl.5.7 of NABL 224.		
5.8	The lab in charge / Lab Engineer shall be authorized to report, review and authorize the test report.		
5.9	Relevant records shall be retained for 1 Year.		
6.0	Requirements For Facilities and Environmental Conditions		
6.1	Facilities and environmental conditions shall be suitable for laboratory activities.		
6.2	Temperature controls shall be provided wherever necessary during the testing of samples (i.e., curing of concrete cube specimens which need to be stored in temperature-controlled water curing tanks as per the Standard Requirement).		
6.3	The temperature of the curing tank shall be monitored and shall be		

	recorded on a daily basis.		
6.4	The laboratory shall have adequate space to carry out all site testing parameters.		
6.5	The laboratory shall be free from any external disturbances that may adversely affect the test results (for example testing should not be conducted where vibrations and other activity affect the test performance).		
6.6	The site laboratory shall be adequately lit, and ventilated with proper water supply arrangements.		
6.7	The lab In-charge/ Lab Engineer shall ensure good housekeeping of the laboratory.		
6.8	Relevant records shall be retained for 1 Year.		
7.0	Test Method Requirements		
	The laboratory shall carry out testing, as specified in the table under Cl.7 of NABL 224.		
8.0	Subcontracting Requirements		
8.1	The recognized laboratory can subcontract the testing activities of their scope of recognition only in case of equipment breakdown/ exigency of work/ unforeseen reasons.		
8.2	The laboratory can subcontract the testing activities to NABL accredited/ recognized laboratory only.		
8.3	The laboratory is responsible for the testing being subcontracted.		
8.4	The subcontracted test results shall be clearly and unambiguously identified in the test report.		
8.5	The laboratory shall have the records for Subcontracting work for selection and approval.		
8.6	Relevant records shall be retained for 1 Year.		
9.0	Purchase Requirements		
9.1	The laboratory shall ensure suitable suppliers are used for the purchase of equipment such as Compressive testing machine (CTM), Electronic weighing balances, and Test sieves, etc. as required for the scope.		
9.2	The laboratory shall have records for all the types of equipment purchased from approved suppliers.		
9.3	The laboratory shall communicate its requirements to suppliers for the purchased items.		
9.4	The laboratory shall maintain the records for the purchased items.		
9.5	The laboratory shall use calibration services from NABL accredited laboratories.		
9.6	Maintenance of the equipment shall be carried out by the laboratory-approved suppliers.		
9.7	Relevant records shall be retained for 1 Year.		

10.0	Equipment Requirements		
10.1	The laboratory shall have the equipment as specified in the table under the Cl. 10 of NABL 224		
10.2	The laboratory shall calibrate the equipment from NABL accredited laboratory as specified in the table under Cl. 10.2 of NABL 224		
10.3	Calibrated equipment shall be labelled, to readily identify the status of calibration of the equipment.		
10.4	The laboratory shall use its own equipment for testing activities.		
10.5	Records of the verifications shall be maintained for cube mould and density cylinders etc.		
10.6	The laboratory shall maintain records for maintenance of equipment used in site testing.		
10.7	The laboratory shall authorize the personnel to operate specific equipment.		
10.8	The laboratory shall verify that equipment conforms to specified requirements before being placed or returned into service.		
10.9	The equipment used for measurement shall be capable of achieving the measurement accuracy required to provide a valid result.		
10.10	Records shall be maintained for all equipment used in the laboratory like purchase, calibration, and maintenance.		
10.11	Relevant records shall be retained for 1 Year.		
11.0	Measurement Traceability Requirements		
11.1	The laboratory shall establish and maintain metrological traceability of its measurement results by means of a documented unbroken chain of calibrations, each contributing to the measurement uncertainty, linking them to an appropriate reference.		
11.2	The laboratory shall ensure that measurement results are traceable to the International System of Units (SI) through the calibration provided by an NABL accredited calibration laboratory.		
11.3	The laboratory shall calibrate all equipment through an NABL accredited calibration laboratory.		
12.0	Sampling Requirements		
12.1	The method of sampling and frequency for testing materials shall be as per the requirements of the project quality plan/ contract agreement.		
12.2	The records must include the relevant details of the sampling information, including identification of the following, where appropriate:		

	• sampling method used • date and time of sampling • unique identification and description the sample (e.g. sample number, amount, name) • identification of the personnel performing sampling • sampling plan • identification of the equipment used • environmental or transport conditions • the location of sampling, including any diagrams, sketches or photographs		
12.3	Relevant records shall be retained for 1 Year.		
13.0	Requirements For Samples (Handling, Storage, Transportation and Disposal)		
13.1	Handling		
	a	Each test item shall be uniquely identified and labelled upon receipt at the laboratory.	
	b	The Laboratory shall not permit any action that may compromise the integrity of the test item	
	c	If identified directly on test sample ensure that the ID tag or other ID method is not likely to come off resulting in a mis-identification or confusion with other samples.	
	d	This unique identification is to be retained with the item throughout the laboratory activities.	
	e	The integrity of the test items is to be maintained throughout their journey from the original location, transportation, receipt, storage, handling, retention, and disposal.	
	f	While handling the sample, the following precautions are to be taken. • To avoid spillage • Keep the sample at appropriate place after use • Avoid any mix up or contamination • Use proper personal protective equipment.	
13.2	Storage		
	a	When test items need to be stored or conditioned under specified environmental conditions, these conditions shall be maintained, monitored and recorded (i.e., concrete cube samples received from site where construction activities are undertaken shall be stored under temperature-controlled water tanks for 7 days and 28 days duration at site laboratory before testing for compressive strength)	
	b	Deviations from specified conditions shall be recorded, asking the customer for further instructions and record the results of the consultation, and inclusion of a disclaimer	
	c	Lab shall retain the tested aggregate samples for minimum period of two weeks unless the requirement is not specified by the customer or the project.	

13.3	Transportation			
	Laboratory is responsible for the transportation of samples from sampling site to the laboratory (if sampling is carried out by the laboratory).			
13.4	Disposal			
	a	The laboratory shall follow appropriate method for disposal of test items, including all provisions necessary to protect the integrity of the test item, and to protect the interests of the laboratory and the customer.		
	b	Disposal of test items to be undertaken with precautions to avoid any damage to environment.		
14.0	Quality Control Requirements			
14.1	The laboratory shall mandatorily participate in at least one product in the Proficiency Testing (PT) program before applying for recognition.			
14.2	The laboratory shall continue participation for its scope in PT programs during the recognition period (minimum one PT program per year covering all the scope).			
14.3	The laboratory shall submit appropriate corrective actions based on root cause analysis in case of non-satisfactory performance in the PT program.			
14.4	The recognized laboratories shall submit the reports of Proficiency Testing (PT) participation annually from the date of issue of the recognition certificate and the same shall be reviewed by NABL.			
14.5	Relevant records shall be retained for 1 Year.			
15.0	Requirements For Test Reports			
15.1	Results obtained from site tests shall be clearly identified on the test Reports, with the details of the following:			
	a	a title "Test Report"		
	b	the location of the performance of the laboratory activities		
	c	identification of the method used		
	d	a description, unambiguous identification, and the condition of the item		
	e	the date of receipt of the testing item(s), the date(s) of performance of the testing activity		
	f	the results with the units of measurement		
	g	deviations, or exclusions from the method.		
	h	identification of the person(s) authorizing the report;		
15.2	The results shall be provided accurately, clearly and unambiguous in a test report.			
15.3	The report shall include all the information agreed with the customer and necessary for the interpretation of the results.			

15.4	All issued test reports shall be retained as technical records for the period of at least 1 year.		
15.5	The results shall be reviewed and authorized prior to release.		
15.6	The lab shall identify the specific test parameter in the test report when the test results are obtained from the Subcontractor.		
16.0	Requirements For Impartiality & Confidentiality		
16.1	Impartiality		
	a Laboratory activities shall be structured and managed to support impartially.		
	b The laboratory shall be responsible for the impartiality of its laboratory activities.		
	c The laboratory shall not allow commercial, financial, or other pressures to compromise impartiality		
	d The laboratory shall be free from any kind of pressure from construction activity (Production, Purchase, or Sales, or any other management).		
	e Risk to impartiality to be reviewed on an ongoing basis.		
16.2	Confidentiality		
	a Personnel, including contractors, personnel of external bodies, or individuals acting on the laboratory's behalf, shall keep confidential all information obtained or created during the performance of laboratory activities, except as required by law.		
	b When the laboratory is required by law or authorized by contractual arrangements to release confidential information, the customer or individual concerned shall, unless prohibited by law, be notified of the information provided.		
17.0	Safety And Regulatory Requirements		
17.1	Proper safety measures during experimentation should be arranged while conducting the tests.		
17.2	Adequate protective clothing, and safety helmets for staff shall be provided. Statutory safety requirements are to be recorded & met.		
17.3	Adequate arrangements for the control of the following shall be available:		
	a Fire safety, Fire alarm.		
	b Shock due to electric current.		
	c Safety guards during load testing.		
17.4	The laboratory shall meet all the relevant regulatory/ statutory requirements (National/ local bodies).		
Date		Signature of the Assessor	

RECOMMENDED SCOPE OF RECOGNITION
(Temporary Site Testing Laboratory)

Name of the Laboratory:	Date(s) of Assessment:
Discipline: Mechanical	Location:

a.	Sl.	Materials or Products tested	Test Parameter	Test Method	Range of Testing	Measurement uncertainty @ Value
	1	Fine Aggregates	Sieve Analysis	IS 2386 (Part-I)		
	2	Fine Aggregates	Bulk Density	IS 2386 (Part-III)		
	3	Fine Aggregates	Specific Gravity	IS 2386 (Part-III)		
	4	Fine Aggregates	Water Absorption	IS 2386 (Part-III)		
	5	Fine Aggregates	Material Finer Than 75 microns	IS 2386 (Part-I)		

b.	Sl.	Materials or Products tested	Test Parameter	Test Method	Range of Testing	Measurement uncertainty @ Value
	1	Coarse Aggregates	Sieve Analysis	IS 2386 (Part-I)		
	2	Coarse Aggregates	Bulk Density	IS 2386 (Part-III)		
	3	Coarse Aggregates	Flakiness index	IS 2386 (Part-I)		
	4	Coarse Aggregates	Elongation Index	IS 2386 (Part-I)		
	5	Coarse Aggregates	Impact Value	IS 2386 (Part-IV)		
	6	Coarse Aggregates	Crushing Value	IS 2386 (Part-IV)		
	7	Coarse Aggregates	Specific Gravity	IS 2386 (Part-III)		
	8	Coarse Aggregates	Water absorption	IS 2386 (Part-III)		
	9	Coarse Aggregates	Material Finer Than 75 microns	IS 2386 (Part-I)		

c.	Sl.	Materials or Products tested	Test Parameter	Test Method	Range of Testing	Measurement uncertainty @ Value
	1	Hardened Concrete	Compressive Strength	IS 516 (Part-1/ Sec-1)		
	2	Fresh Concrete	Workability by Slump	IS 1199 (Part-2)		

Signature, Date & Name of lab Representative	Signature, Date & Name of Assessor(s)	Signature, Date & Name of Lead Assessor
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DECLARATION OF IMPARTIALITY & CONFIDENTIALITY
(to be filled in by each Assessor and enclosed with the Assessment report)

Name		Assessor ID: (To be filled in by NABL Sect.)
Designation		
Organisation		
Address		
Capacity	Lead Assessor / Technical Assessor / Technical Expert / Observer	
Lab Assessed		
Date of Assessment		
Type of Assessment	<i>Initial assessment / Onsite Surveillance / Supplementary visit</i>	

I _____, hereby declare that I have not offered any consultancy, guidance, supervision or other services to the lab (e.g. internal audit), in any way.

I am / am not* an ex-employee of the lab and am/ am not* related to any person of the management of the lab.

I got an opportunity to go through various documents like Quality Manual, Procedural Manuals, Work instructions, Internal reports etc. of the above lab and other related information that might have been given by NABL. I undertake to maintain strict confidentiality of the information acquired in course of discharge of my responsibility and shall not disclose to any person other than that required by NABL.

Date:

Place:

Signature

OBSERVATION SHEET

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